



## Test 1 FSPM0024 20212022-3

CALCULUS - foundation (Universiti Teknologi Malaysia)



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**Foundation Programme**  
**Universiti Teknologi Malaysia**  
**Session 2021/2022 Semester 3**

*Test 1 (15%)*

**FPSM 0024 Calculus**

*Answer all SIX questions. Show all your work clearly. Time given is 1 hour and 15 minutes with additional 15 minutes for submission.*

1. Given function  $h(x)$  such that

$$h(x) = \begin{cases} 6 - kx, & x \leq 2 \\ x^2 - 10, & 2 < x \leq 4 \\ \frac{12}{x-2}, & x > 4 \end{cases}$$

i. Find  $k$  if  $\lim_{x \rightarrow 2} h(x)$  exists.

(2 marks)

ii. Determine whether  $h(x)$  is continuous at  $x = 4$ .

(3 marks)

2. Let  $f(x) = \frac{x-1}{x^2-1}$ . Find

$$\lim_{x \rightarrow \infty} xf(x) + \lim_{x \rightarrow 1} \frac{x}{f(x)}.$$

(4 marks)

3. Use the First Principle to find the derivative of  $f(x) = \frac{1}{2\sqrt{x}+1}$ .

(4 marks)

4. Find  $\frac{dy}{dx}$  for the curve given as  $ye^{x+y} = \sin(3x - y^2)$  by using implicit differentiation.

(5 marks)

5. Given a set of parametric equations

$$x = \ln(1 + t^2) \quad \text{and} \quad y = 2t^3 - 2t^2.$$

i. Find  $\frac{dy}{dx}$  in terms of  $t$ .

(3 marks)

ii. Find  $\frac{d^2y}{dx^2}$  at  $t = 1$ .

(3 marks)

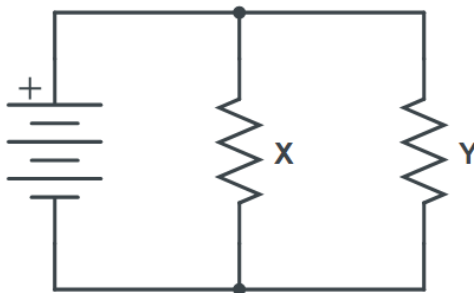


Figure 1: Figure for Question 6

6. Figure 1 shows a circuit where  $X$  and  $Y$  are the resistors. The total resistance  $R$ , is given by

$$\frac{1}{R} = \frac{1}{X} + \frac{1}{Y}.$$

The resistances of  $X$  and  $Y$  are 30 ohms and 50 ohms respectively, and the resistances increase over time.

i. Write an equation of related rate of change involving  $\frac{dR}{dt}$ ,  $\frac{dX}{dt}$  and  $\frac{dY}{dt}$ .

(3 marks)

ii. If  $\frac{dX}{dt} = 0.03$  ohm/hour and  $\frac{dY}{dt} = 0.02$  ohm/hour, find  $\frac{dR}{dt}$ .

(3 marks)

**END OF QUESTIONS**

## LIST OF FORMULA

| Trigonometric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$ |
| Logarithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Differentiations                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{d}{dx}[k] = 0, k \text{ constant}$                                                                                                 |
| $\frac{d}{dx}[x^n] = nx^{n-1}$                                                                                                            |
| $\frac{d}{dx}[e^x] = e^x$                                                                                                                 |
| $\frac{d}{dx}[\ln  x ] = \frac{1}{x}$                                                                                                     |
| $\frac{d}{dx}[\cos x] = -\sin x$                                                                                                          |
| $\frac{d}{dx}[\sin x] = \cos x$                                                                                                           |
| $\frac{d}{dx}[\tan x] = \sec^2 x$                                                                                                         |
| $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$                                                                                        |
| $\frac{d}{dx}[\sec x] = \sec x \tan x$                                                                                                    |
| $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$                                                                   |
| Parametric Differentiation                                                                                                                |
| $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ $\frac{d^2y}{dx^2} = \frac{d}{dt} \left( \frac{dy}{dx} \right) \times \frac{dt}{dx}$ |